pitfalls. On July 1, the Union cabinet approved a Rs 1 lakh crore Research Development Innovation (RDI) scheme—a bold move to transform private sector R&D, first announced in the July 2024 Union budget. Sanjeev Bikhchandani, founder and executive vice chairman of Info Edge, believes it could be a "game changer". "India must invest in deep tech—not just for economic growth but as a strategic imperative," he says.

Drafting India's largest-ever R&D stimulus meant designing new financial instruments with enough flexibility to attract private capital. It took nearly a year to complete this groundwork. Progress was slowed by inter-departmental turf battles, while private players raised concerns over access criteria, IP ownership and a perceived tilt toward academia. So, the scheme explicitly excludes universities.

The fund will be routed through a two-tier mechanism. At the first level, there will be a Special Purpose Fund (SPF) within the Department of Science & Technology's (DST) Anusandhan National Research Foundation. From the SPF, funds will be allocated to second-tier professional fund managers. They will evaluate proposals, prioritise high-readiness projects, and deploy capital as long-term concessional loans, equity for startups or tech acquisition support, not just grants. As Rahul Walawalkar, president of NETRA (Net-Zero Energy Transition Association), puts it: "Technologies must be ready to scale, bank and deploy, not sit in labs."

Execution will be everything. India lacks an ecosystem of independent tech evaluators, validation labs and public testbeds. Patent processing can take 3–5 years. Lab equipment remains costly due to high import duties. The bureaucratic culture may be harder to fix. "This scheme will work only if fund managers are empowered to make bold calls and resist L1 (lowest bidder) tender logic and audit fears," says a top DST official. Ashok Atluri, managing director of Zen Technologies, agrees: "You will have to dig deeper, understand the pain of the innovator and back his instinct."

WHY THE RDI SCHEME IS CRUCIAL

- **India's R&D spend is below 0.7% of GDP; private sector share is just 36%—well below global benchmarks**
- **The fund targets high-risk, late-stage technologies where most Indian prototypes stall**
- **It prioritises sovereign IP in strategic sectors like semiconductors, clean energy and quantum tech**
- **Companies can use it to acquire and license cutting-edge global technologies**
- **Designed to avoid past failures. Will use professional fund managers, eye commercial deployment**

PROTOTYPE TO DEPLOYMENT

For decades, India's innovation model has veered between two extremes: underfunded public science and a private sector focused on near-term services or low-risk incremental products. The result has been a culture of *jugaad*, not foundational invention. Total R&D spending remains stuck below 0.7 per cent of GDP, while China's exceeds 2.4 per cent and the US's is over 3.5 per cent. Private sector contribution to India's total R&D expenditure is just 36 per cent—a far cry from the 70–80 per cent in advanced economies. "Globally, governments back their scientists," says Ajai Chowdhry, co-founder of HCL. "They give challenges to the private sector with the assurance: if you solve this, we'll buy it. In India, innovators don't know who will buy their product."

WHO WILL GET FUNDED

Models abroad offer useful cues. The US's Defense Advanced Research Projects Agency grants full autonomy to elite programme managers—often from academia or industry—who lead high-risk projects with clear timelines and accountability. This has delivered breakthroughs like GPS, the Internet and stealth aircraft. Citing clean tech, Jagjeet Sareen of Dalberg Advisors says India's R&D needs more than capital; it needs ecosystem thinking, shared infrastructure and stronger focus on commercialisation to "meet climate goals through indigenous innovation".

The government says the fund will focus on strategic tech sectors where the country has either latent capability or urgent need: quantum tech, semiconductors, clean energy, AI, biotech, space and advanced materials. Around 50–60 per cent will be deployed as low- or zero-interest loans to private firms undertaking high-risk, late-stage R&D. Another slice will support co-development consortia combining startups, academia and large companies where India aims to build sovereign IP. A smaller but critical portion is set aside for global technology acquisition, a departure from the old import substitution model.

Several industry players whom INDIA TODAY spoke to stress the need to avoid 'spray and pray' funding. Instead, the fund should nurture focused clusters and support talent pipelines by funding postdoctoral and R&D fellowships in the private sector. If it works, the RDI scheme could finally end India's *jugaad* era and seed a new generation of original inventors. ■